Computer Vision Practical Report:

TOP HAT TECHNIQUE

1. AIM

To compute the Top-Hat transform (the difference between the original image and its morphological opening) to isolate elements and structures that are brighter than their surrounding environment using cv2.morphologyEx() in OpenCV, display the results, and save the output image.

2. PROCEDURE

- Load the input image using cv2.imread() and convert it to grayscale using cv2.cvtColor().
- Define a spatial structuring element (kernel) using cv2.getStructuringElement() (e.g., rectangular or elliptical).
- Apply the Top-Hat transform using cv2.morphologyEx() with cv2.MORPH_TOPHAT.
- Display the original and Top-Hat transformed images side-by-side using Matplotlib.
- Save the output image to disk using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load image
img_bgr = cv2.imread('E34-IMG.jpg')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# 2. Define Structuring Element (9x9 kernel to isolate small bright spots)
kernel = np.ones((9, 9), np.uint8)

# 3. Perform Top Hat Operation (Original - Opening)
tophat = cv2.morphologyEx(img_rgb, cv2.MORPH_TOPHAT, kernel)
```

```python
# 4. Display side-by-side on a SINGLE PAGE
fig, axes = plt.subplots(1, 2, figsize=(8, 4))

axes[0].imshow(img_rgb)
axes[0].set_title('Original Image')
axes[0].axis('off')

axes[1].imshow(tophat)
axes[1].set_title('Top Hat Image')
axes[1].axis('off')

plt.tight_layout()
plt.show()
```

4. OUTPUT

5. RESULT

The Python script executed successfully. The input image was loaded.